

EMCS Design Review Meeting WebNSV

13th September 2023

NEC Corporation

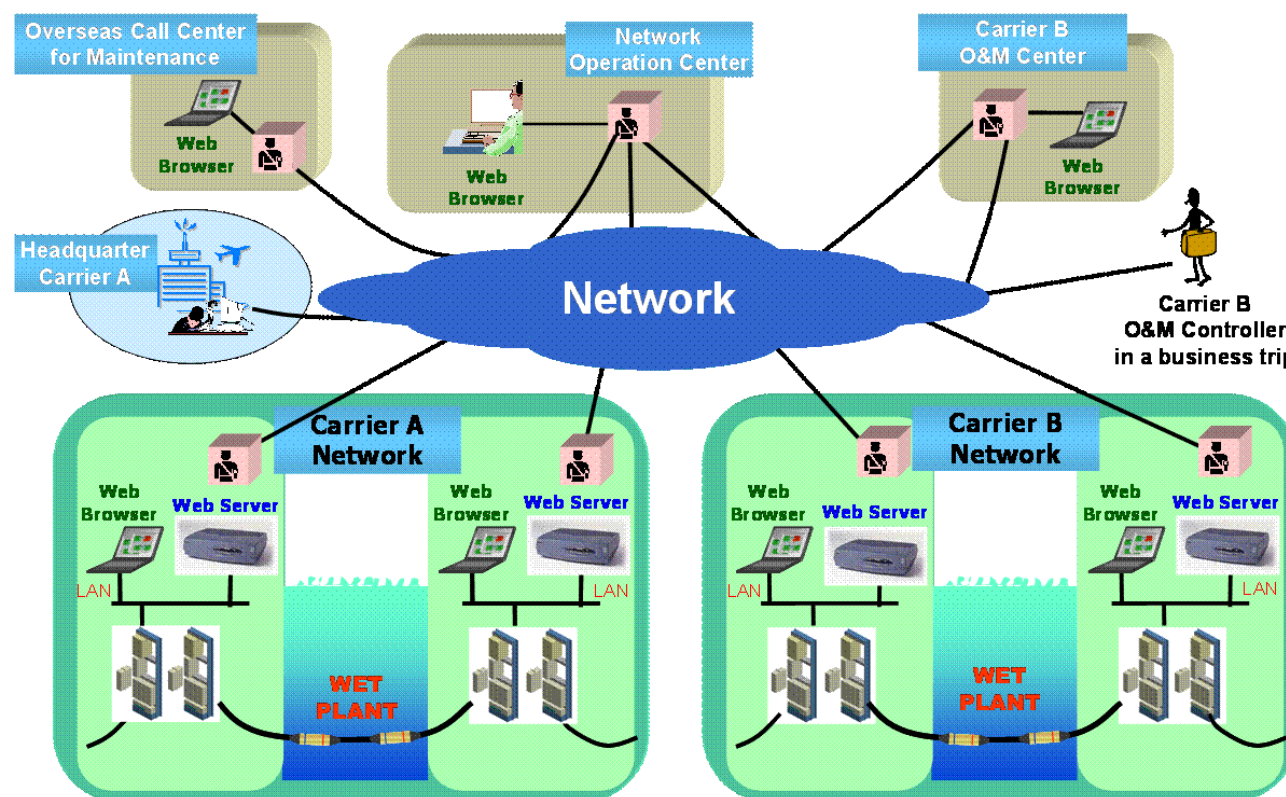
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1. Introduction

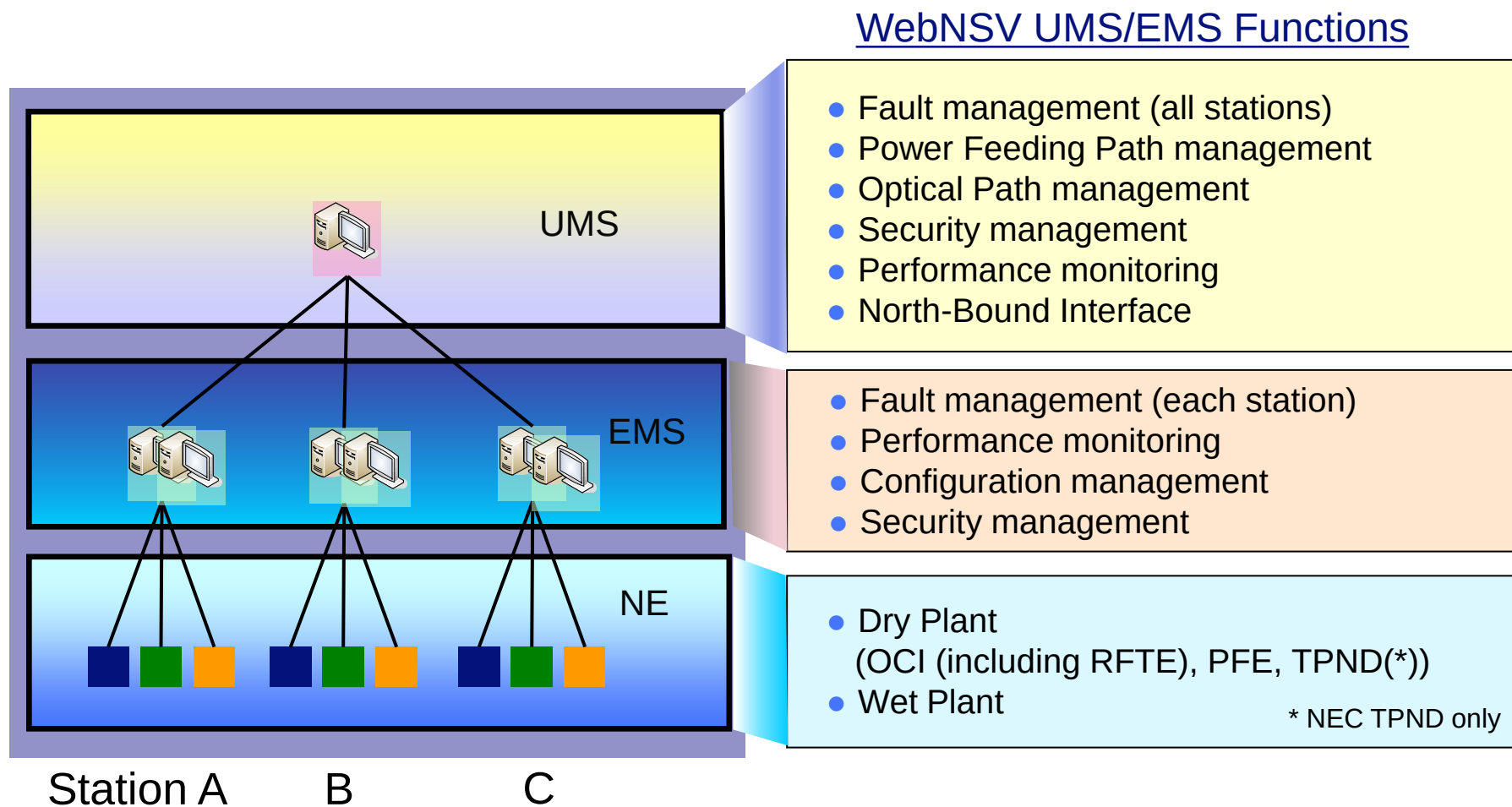
1. Introduction

- ◆ WebNSV provides all the functions required for submarine network management, which are Performance Management, Fault Management, Configuration Management, Security Management, and Optical Path Management.



1. Introduction

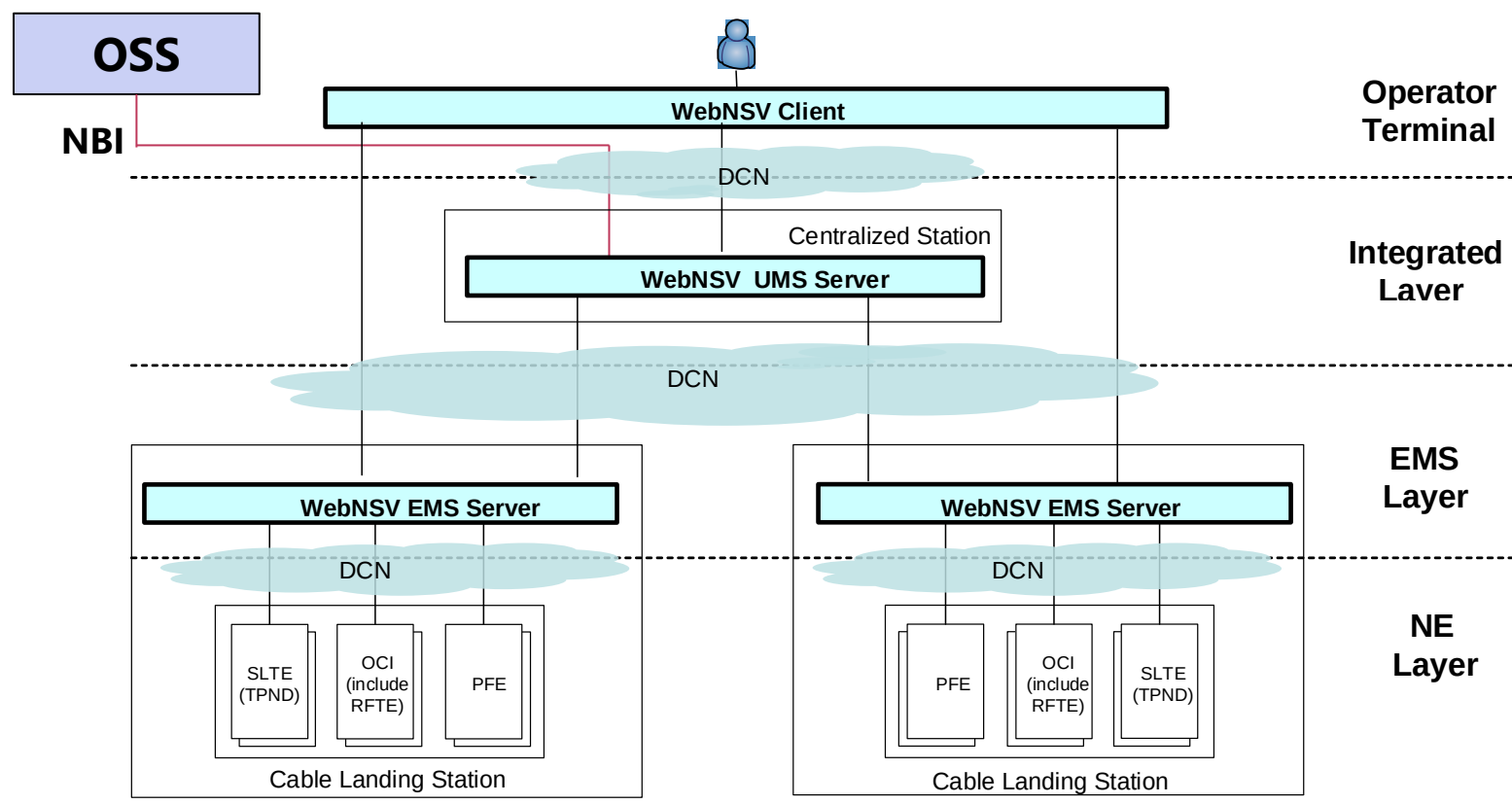
- ◆ WebNSV EMS/UMS supports both network management and element management layers for submarine segment.



2. Architecture

2. Architecture

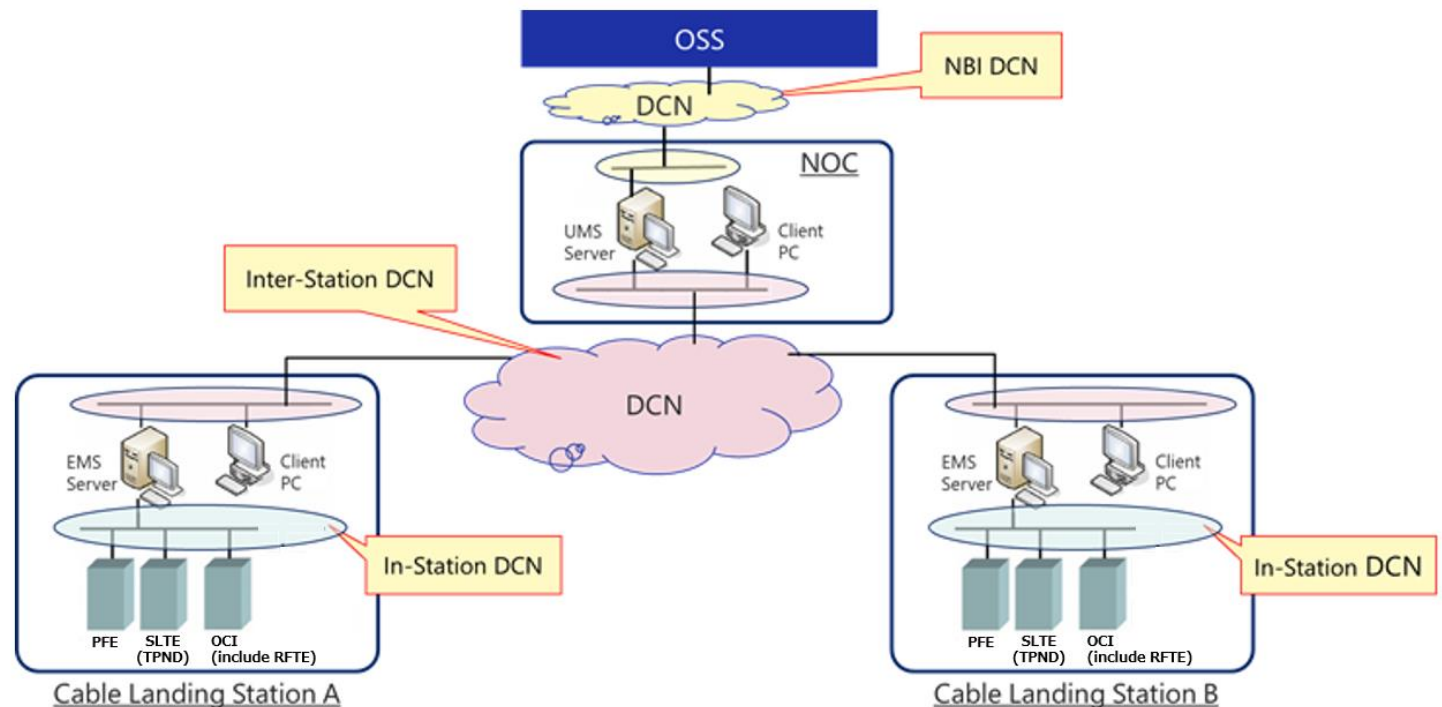
- ◆ WebNSV applies the distributed architectures that consists of the Server (WebNSV EMS server and WebNSV UMS server) and the Client (WebNSV client).
- ◆ Client application is distributed from Server using Java Getdown technology.



WebNSV available structures

2. Architecture

- ◆ The WebNSV EMS is installed in each cable landing station.
- ◆ The WebNSV UMS is normally installed in the centralized station.
- ◆ The WebNSV Client can be commonly used for both types of servers as long as their communication links are available through DCN.
- ◆ The WebNSV UMS provides North-Bound interface.



2. Architecture

◆ WebNSV EMS/UMS configuration in EMCS

■ EMS:

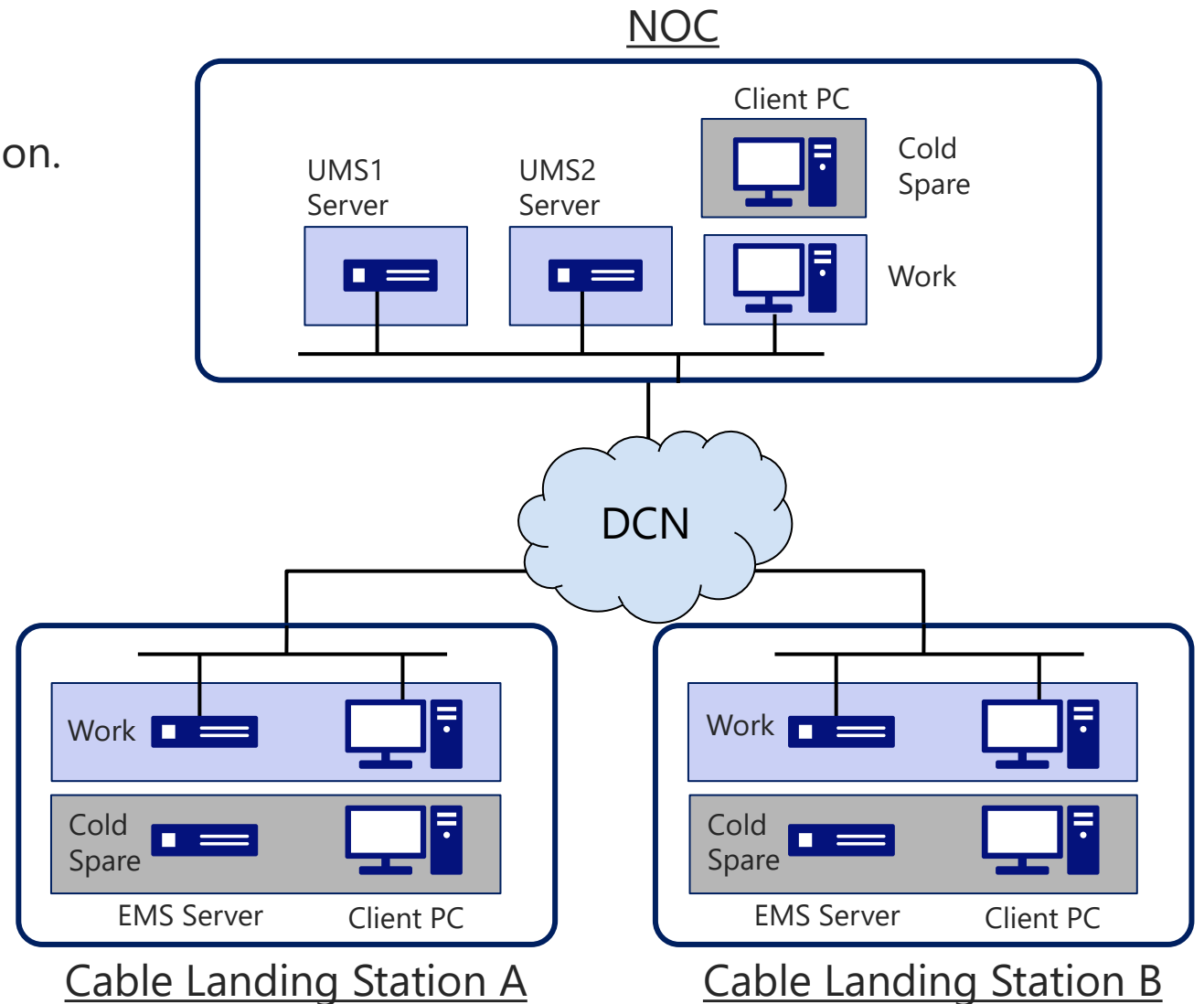
- ✓ One EMS server with one spare at each station.
- ✓ Cold standby with iLO.

■ UMS:

- ✓ Two UMS server at NOC.
- ✓ Work/Work.

■ Each server :

- ✓ Redundant by RAID configuration.



3. Key features of WebNSV EMS/UMS

3. Key features of WebNSV EMS/UMS

	WebNSV EMS	WebNSV UMS
Fault management	➤ Fault detection and localization of NEs in each station ➤ Fault detection and localization of Wet Plant	➤ Fault detection and localization of NEs for all stations ➤ Fault detection and localization of Wet Plant
Performance management	➤ Performance monitor of NEs in each station ➤ Performance monitor of Wet Plant	➤ Performance monitor of Wet Plant
Configuration management	➤ Parameter setting of NE	➤ Power Feeding Path management ➤ Power Feeding Support ➤ Optical SW management (not apply to EMCS) ➤ ROADM unit management (not apply to EMCS)
Log management	➤ Log of event data and performance data of NEs ➤ Log data back up	➤ Log of event data of stations in network ➤ Log data back up
Security management	➤ Access control of user ➤ Security Setting with FP unit	➤ Access control of user ➤ Security Setting with FP unit and Function unit
NBI	N/A	➤ Alarm information ➤ Performance monitor information ➤ Inventory information

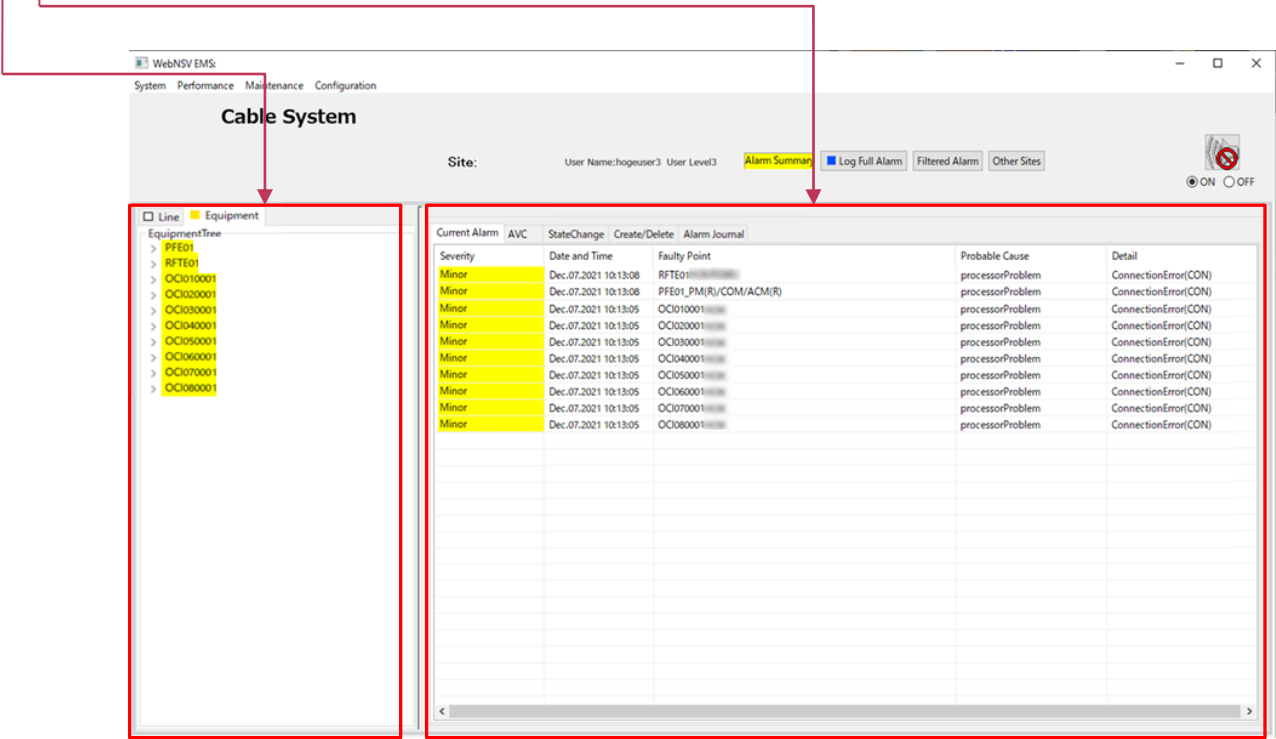
4. WebNSV EMS function

- 4.1 EMS Fault management
- 4.2 EMS Performance management
- 4.3 EMS Configuration management
- 4.4 EMS Log management
- 4.5 EMS Security management

4.1 EMS Fault management

◆ Fault detection and localization of NEs in each station

- NE that detects any failure is indicated in Top window.
- Detail alarms are displayed in journal window of Top window.



Example of Top window

Object Color	Alarm Severity
Dark Red	Critical
RED	Major
Yellow	Minor
Light Yellow	Warning
Green	Indeterminate
Blue	Cleared
Grey	Unknown

4.1 EMS Fault management

◆ Fault detection and localization of Wet Plant by using OCI/RFTE.

■ Repeater Section Monitoring:

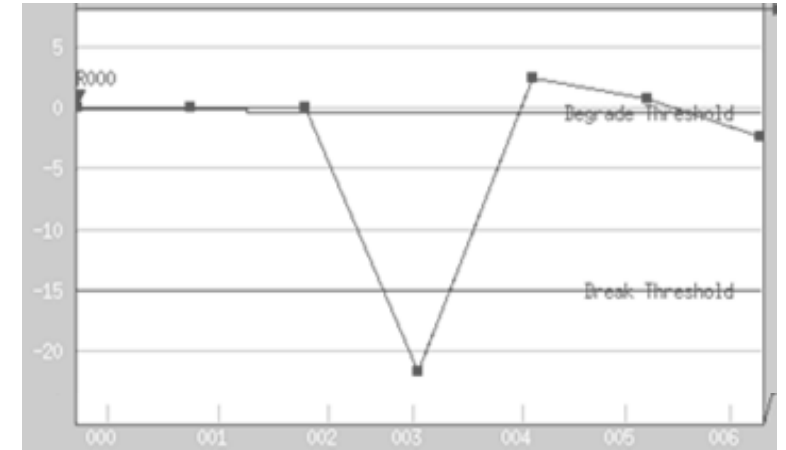
- Operation of periodic Wet Plant monitoring and individual measurement.
- Fault analysis and detection
- Visualize of Repeater Section gain graph for detailed analysis.

■ Cable Monitoring:

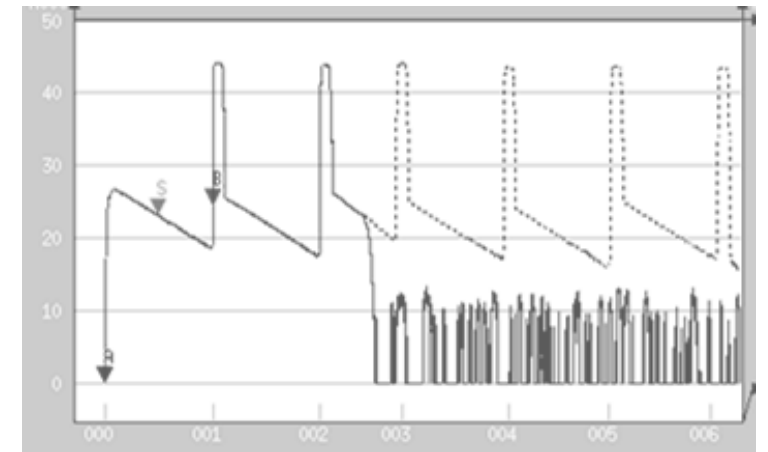
- Control of cable trace measurement.
- Visualize of measurement cable trace.

➤ Example of Wet Plant fault;

- Repeater Power Reduction (including ROADM)
- Cable span loss increase



Example of Repeater section gain



Example of Cable loss trace

(Note) GUI design is subject to change in actual software.

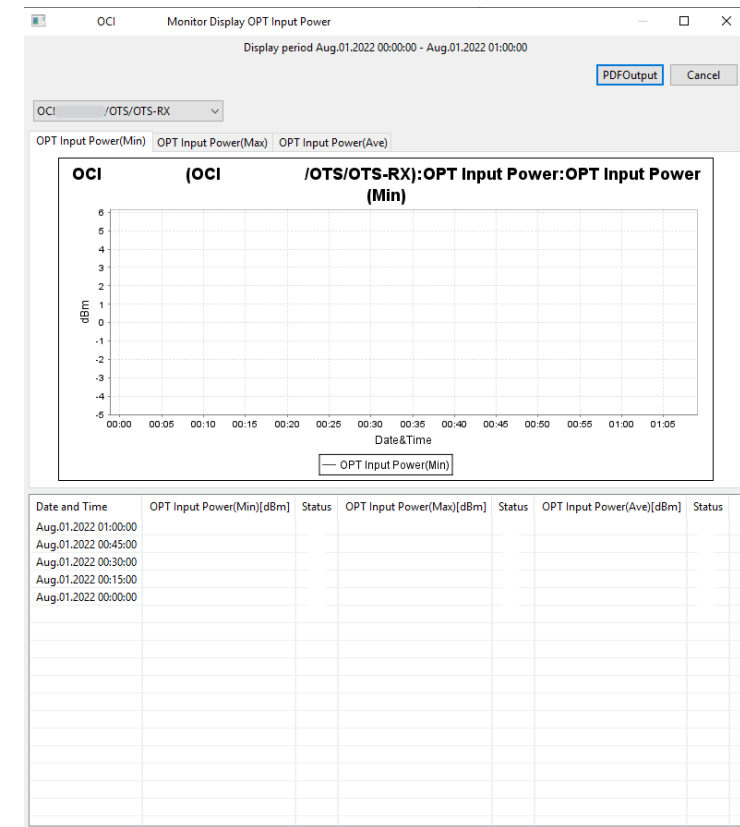
4.2 EMS Performance management

◆ Performance monitor of NEs in each station

- EMS provides the Dry Plant performance monitor function in the station.
- There are two types of monitor data: on demand acquiring and periodically reported.
- PM of transmission quality is periodically reported from each NEs.
- These data is displayed in both graphical and tabular format in PM display window.

◆ Performance monitor of Wet plant

- EMS provides the Wet Plant performance monitor measured by OCI/RFTE.
- Example of performance monitor (Wet Plant);
 - ✓ Deviation of Repeater Section Gain
 - ✓ Repeater Power Reduction
 - ✓ Cable span loss increase
 - ✓ Gain tilt change

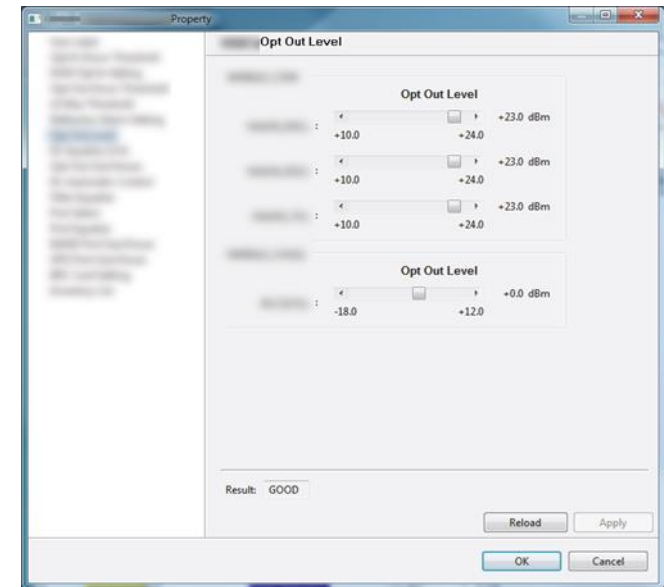
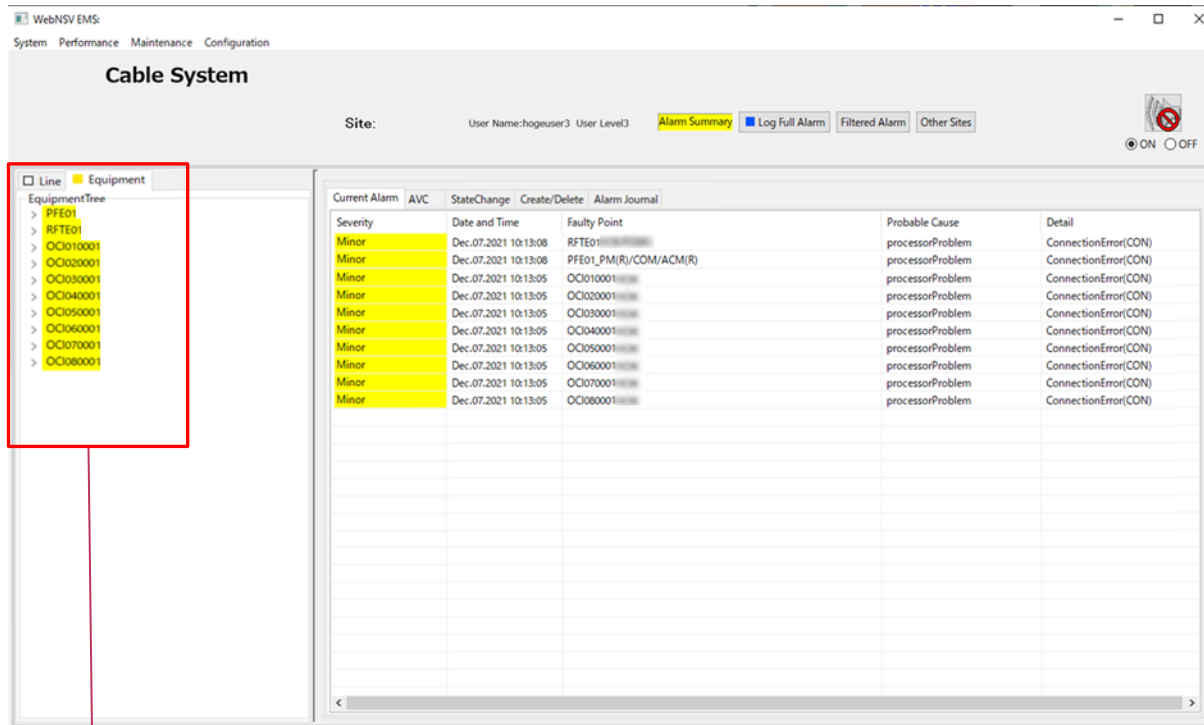


Example of PM display window

4.3 EMS Configuration management

◆ Parameter Setting of NE

- NE's parameter is configured and is displayed in sub-window of Top window.



Example of Property window

Select NE and open Property window.

4.4 EMS Log management

- ◆ Log of event data and performance data of NEs
 - Log of event data is displayed in Log display window.
 - Log of PM data is displayed in PM display window.
- ◆ Log data back up
 - Log data can be back up to external device.

The screenshot shows a 'Log Display' window with the following components:

- Time Range:** Two date pickers for 'From' and 'To'. 'From' is set to Jan 1, 2022. 'To' is set to Aug 10, 2022.
- NE Selection:** A dropdown menu set to 'ALL'.
- Alarm:** A tabbed interface with 'Alarm' selected. Other tabs include 'A/C', 'StateChange', 'Create/Delete', 'User Security', and 'All'.
- 'Severity' Filter:** Checkboxes for 'Cleared', 'Indeterminate', 'No Alarm', 'Warning', 'Minor', 'Major', 'Critical', 'Mask ON', and 'Mask OFF'. 'Cleared' and 'Indeterminate' are checked.
- 'Probable Cause' Filter:** An empty text input field.
- 'Detail' Filter:** An empty text input field.
- Buttons:** 'Operate Display' and 'PDF Report'.
- Jump to Correlation:** A button located below the filters.
- Data Table:** A table with 7 columns: Severity, Date and Time, Faulty Point, Probable Cause, Detail, Notification ID, and Correlated ID. The table is currently empty.
- Page Info:** 'page: 1 / 1' and navigation buttons '<< Back', 'Next >>', and 'Cancel'.

Example of Log display window

4.5 EMS Security management

◆ Access control of user

■ EMS provides three authority levels for user security.

- The highest level of user can monitor all information, configure NEs, operate EMS system function such as Database backup, and so on.
- The lowest level of user can monitor some information.

■ Login user is authenticated with password.

- User ID and Password is stored in EMS Database.

■ These settings of user security are set in User information window.

Tasks	User Level 1	User Level 2	User Level 3
Retrieving NE Attribute*	✓	✓	✓
Retrieving Log Record	✓	✓	✓
Retrieving PM Data	✓	✓	✓
Retrieving Alarm Status	✓	✓	✓
Setting NE Attribute*	—	✓	✓
User Management	—	—	✓
Login Management	—	—	✓
Deleting Log Record	—	—	✓
Creating/Downloading CSV File	—	—	✓
Database Backup	—	—	✓
Downloading Trace/Access Log	—	—	✓

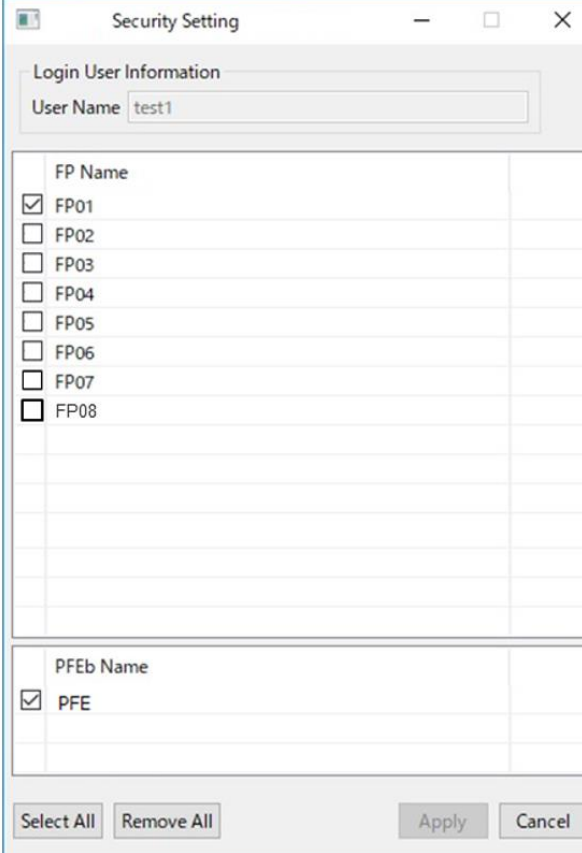
* NE Attribute shows an object NE Status and is the information retrieved/set from/to NE.

Example of User information window

4.5 EMS Security management

◆ Security Setting with FP unit

- EMS provides FP unit security. These settings are set in Security Setting window, and can be set to each user individually.
- When the FP is selected by checkbox, the NEs only relating to the FP are displayed on EMS window to the user.
- PFE is commonly used regardless of FP. Therefore, PFE is directly selected by checkbox.



The screenshot shows a 'Security Setting' window with the following components:

- Login User Information:** A text field labeled 'User Name' containing the value 'test1'.
- FP Name:** A table with 8 rows, each containing a checkbox and a label (FP01 through FP08). The checkbox for FP01 is checked.
- PFEb Name:** A table with 1 row containing a checked checkbox and the label 'PFE'.
- Buttons:** 'Select All', 'Remove All', 'Apply', and 'Cancel' are located at the bottom right of the window.

Login User Information	
User Name	test1

FP Name	
☒	FP01
☐	FP02
☐	FP03
☐	FP04
☐	FP05
☐	FP06
☐	FP07
☐	FP08

PFEb Name	
☒	PFE

Select All Remove All Apply Cancel

Example of Security Setting window

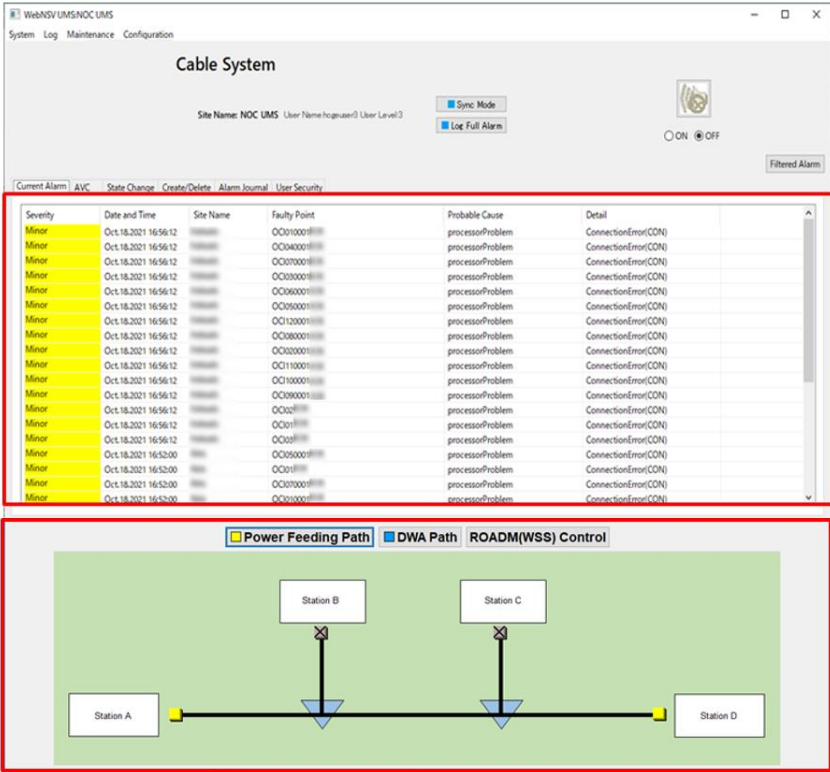
5. WebNSV UMS function

- 5.1 UMS Fault management
- 5.2 UMS Performance management
- 5.3 UMS Configuration management
- 5.4 UMS Log management
- 5.5 UMS Security management
- 5.6 UMS NBI (North-Bound Interface)

5.1 UMS Fault management

◆ Fault detection and localization of NEs for all stations

- Station that detects any failure is indicated in Top Window.
- Detail alarms are displayed in journal area.



Object Color	Alarm Severity
Dark Red	Critical
RED	Major
Yellow	Minor
Light Yellow	Warning
Green	Indeterminate
Blue	Cleared
Grey	Unknown

Example of Top Window

5.2 UMS Performance management

◆ Performance monitor of Wet Plant

- UMS provides the Wet Plant performance monitor function.
- Wet Plant performance monitor data can be displayed on UMS.

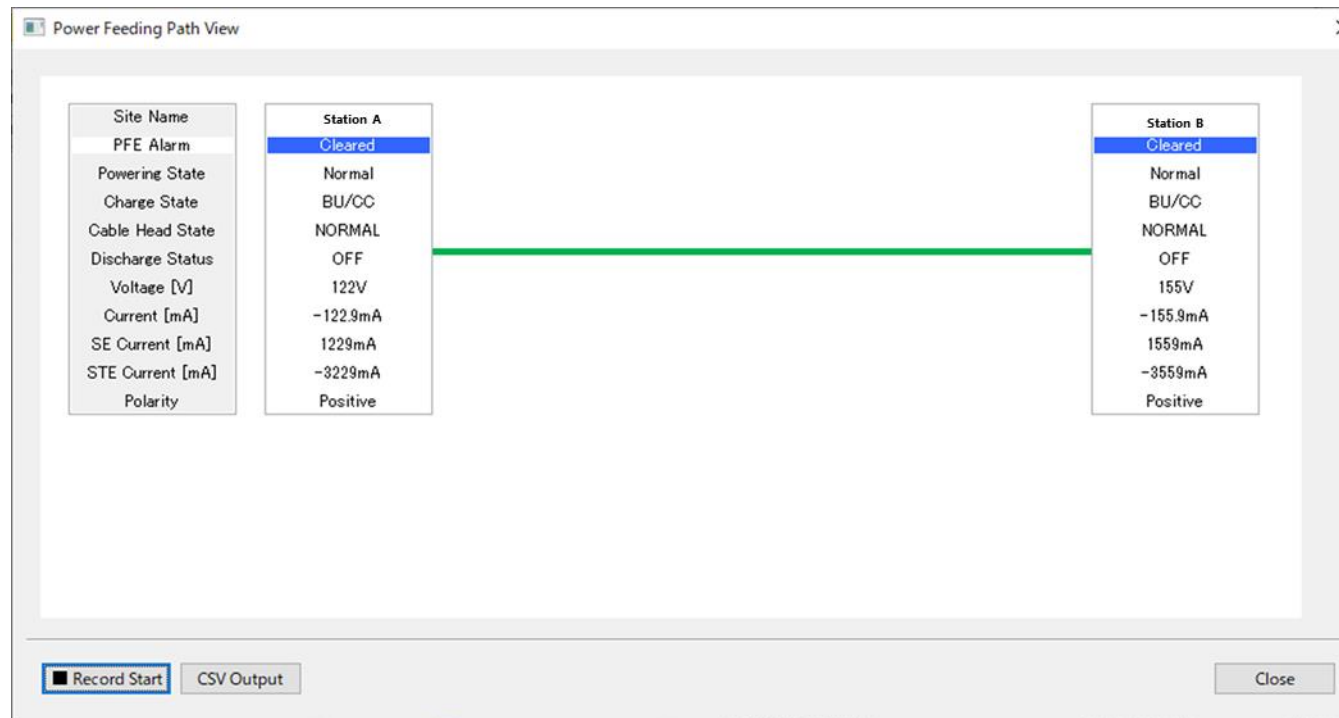
➤ Example of performance monitor of Wet Plant ;

- ✓ Deviation of Repeater Section Gain
- ✓ Repeater Power Reduction
- ✓ Cable span loss increase
- ✓ Gain tilt change

5.3 UMS Configuration management

◆ Power Feeding Path management

- UMS provides the Power feeding path management function.
- UMS has “Power Feeding Path View” display which enables to monitor the system powering status with all PFE’s operating conditions.



Example of Power Feeding Path View window

5.3 UMS Configuration management

◆ Power Feeding Support function

- UMS provides power feeding support function which shows the operating procedure of power feeding path switching.

The screenshot displays the 'Power Feeding Support(PIC)' window. At the top, it shows configuration details: Fault Mode(Shunt/Open): Shunt Fault, Fault Mode(Fiber): without Fiber Break, Fault Position: S1, and Support Pattern: During Repair. Below this, it indicates Scenario Version: 0.1 and Scenario Updated Date: 2021/11/09 10:00:00. A 'View Only User: offline' status is shown. The main area contains a table with columns for Step, Operation, PIC Logged in User, Expected Value, Station A status, Station B status, and PIC Final Check. The table lists three steps: 0001 (No operation is required), 0002 (Power off (NOR/CC) StationA PFE), and 0003 (Power off (NOR/CC) StationB PFE). Each step has corresponding actions and status indicators (Completed, Next, No Action).

Step	Operation	PIC Logged in User	Expected Value	Station A	Station B	PIC Final Check
		hogeuser3	(Expected PFE Voltage with +/- 500V accuracy) (Expected PFE Ampere with +/- 5% accuracy)	offline	offline	
0001	No operation is required.	Completed		No Action	No Action	Next
0002	Power off (NOR/CC) StationA PFE	Completed	StationA:0mA	No Action	Press STOP button	Next
0003	Power off (NOR/CC) StationB PFE		StationB:0mA	Press STOP button	No Action	

Example of Power Feeding Support window

5.3 UMS Configuration management

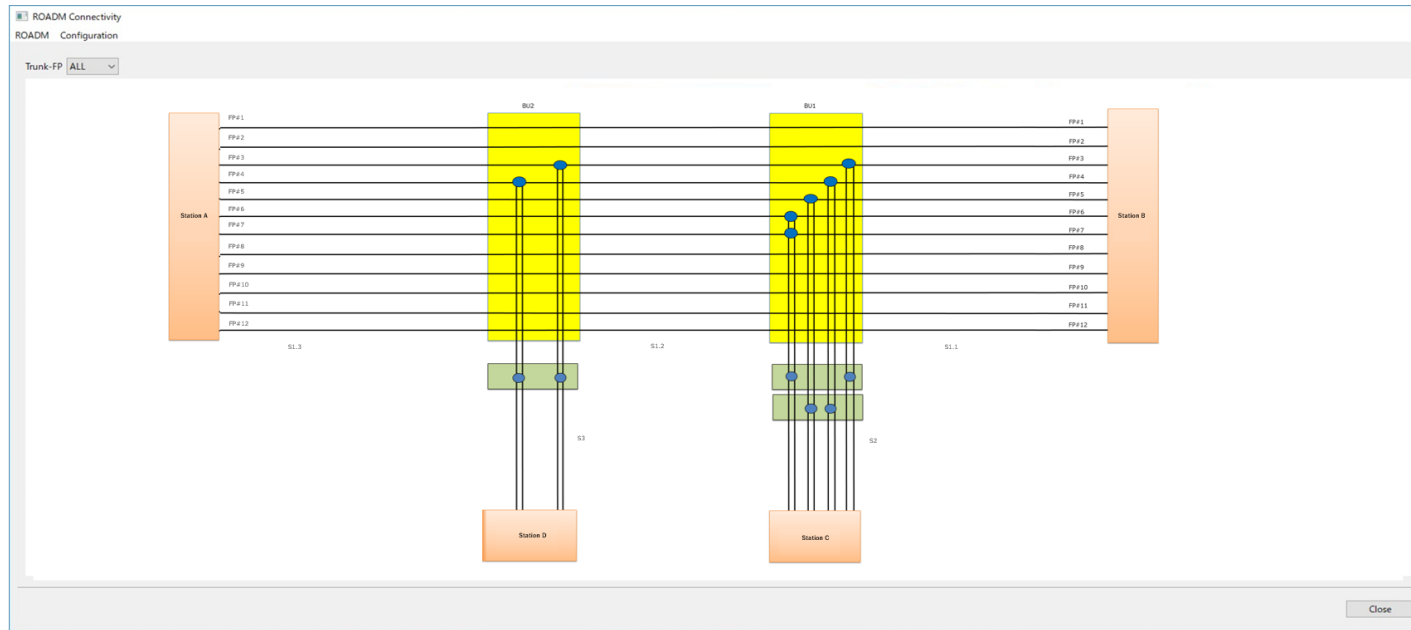
◆ Optical SW management (not apply to EMCS)

- UMS provides the optical SW management function of PSBU.
 - The status of the optical SW in PSBU can be changed by UMS operation.
 - Branch Through or Trunk Through for PSBU.
 - The present status of the optical SW can be displayed on UMS window.

5.3 UMS Configuration management

◆ ROADM unit management (not apply to EMCS)

- UMS provide the ROADM unit management function.
- The status of ROADM unit are displayed in ROADM Connectivity window.



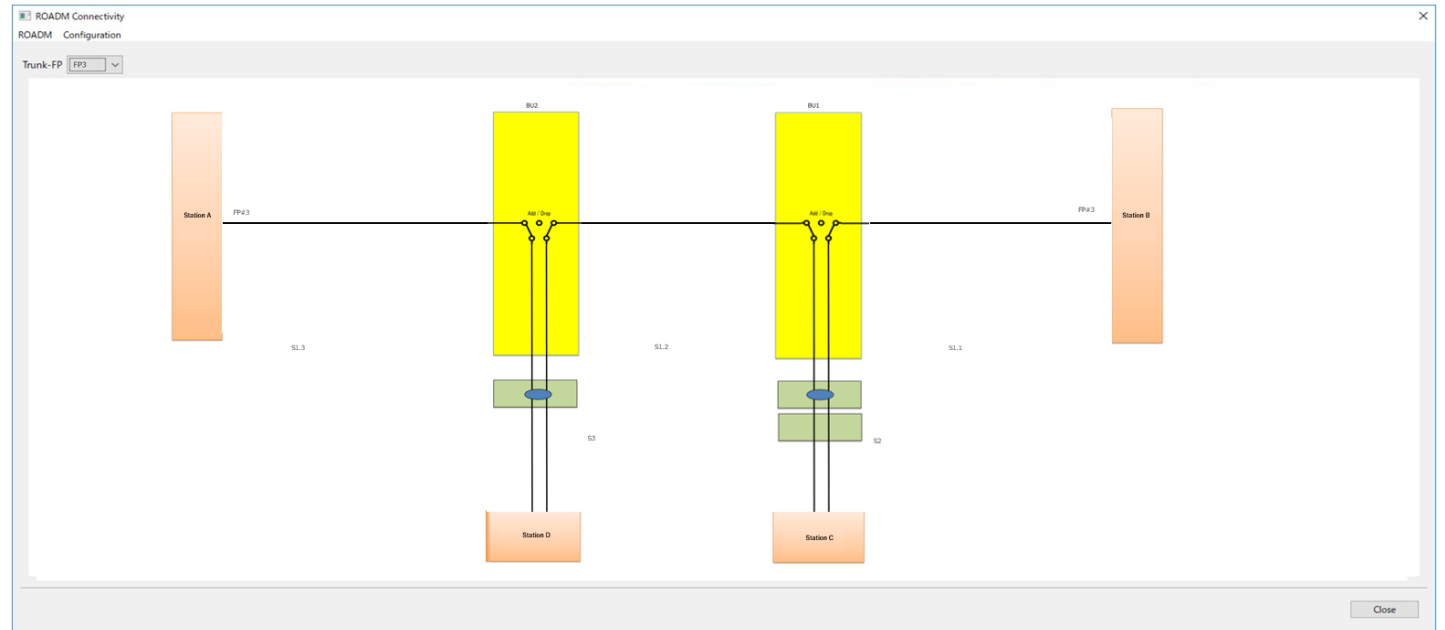
Example of ROADM connectivity window (FP=ALL)

5.3 UMS Configuration management

◆ ROADM unit management (not apply to EMCS)

■ The following functions to control ROADM are provided.

- ✓ Add/Drop Setting
- ✓ Level adjustment Setting
- ✓ OSM (Optical Spectrum Monitor)

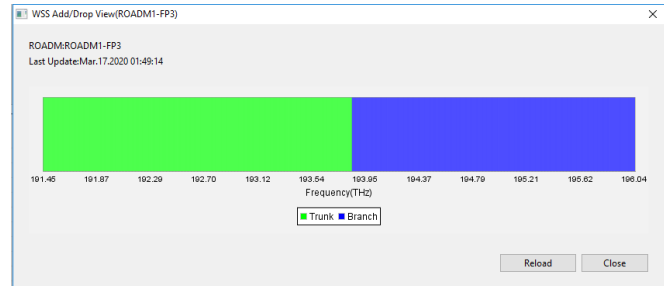


Example of ROADM connectivity window (Selected FP)

5.3 UMS Configuration management

- ◆ ROADM Add/Drop Setting (not apply to EMCS)

- Add/Drop Setting is set by specifying the frequency range and the direction (Add/Drop or Trunk Through).
- The present status of Add/Drop Setting can be displayed graphically.



Example of ROADM Add/Drop view window

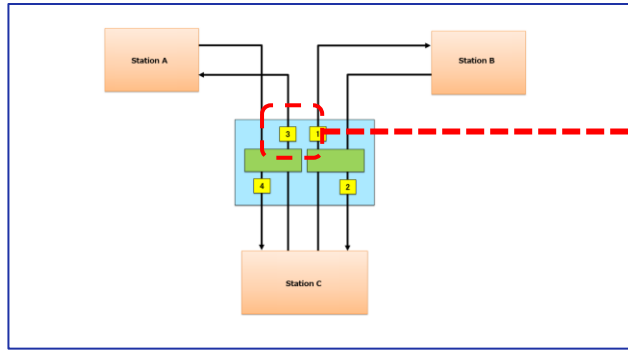
[illegible]

Example of ROADM Add/Drop setting window

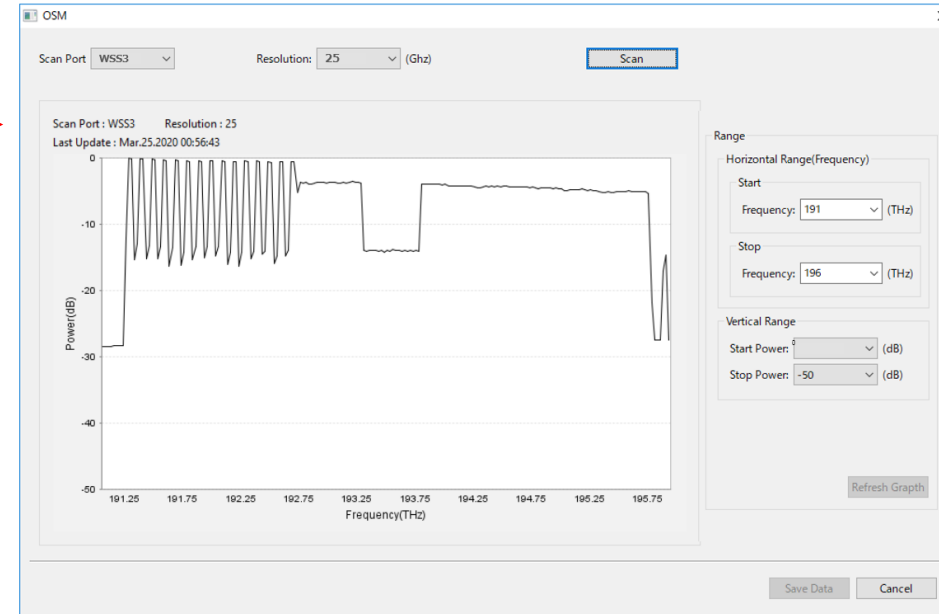
5.3 UMS Configuration management

◆ ROADM OSM (not apply to EMCS)

- OSM function can be executed by specifying the target port, the resolution, the frequency range and the power range.
- The monitored spectrum can be displayed graphically.



Example of Level adjustment / OSM window



Example of OSM window

5.4 UMS Log management

- ◆ Log of event data of stations in network
 - Log of event data is displayed in Log display window.
- ◆ Log data back up
 - Log data can backup to external device.

The screenshot shows the 'NOC UMS Log Display' window. It features a 'Time Range' section with 'From' and 'To' date pickers. The 'From' date is Jan 1, 2022, and the 'To' date is Aug 10, 2022. Below this are dropdowns for 'Site' (ALL) and 'NE' (ALL). A tabbed interface shows 'Alarm' selected, with other tabs being 'AVC', 'State Change', 'Create/Delete', 'User Security', and 'ALL'. The 'Severity' filter includes checkboxes for Cleared, Indeterminate, No Alarm, Warning, Minor, Major, Critical, Mask ON, and Mask OFF. There are also input fields for 'Probable Cause' and 'Detail' filters, and a 'Display' button. A 'PDF Report' button is also present. A 'Jump To Correlation' button is located above the data table. The table has columns for Severity, Date and Time, Site Name, Faulty Point, Probable Cause, Detail, Notification ID, and Correlated ID. At the bottom, there are navigation buttons: '<< Back', 'Next >>', and 'Cancel'.

Severity	Date and Time	Site Name	Faulty Point	Probable Cause	Detail	Notification ID	Correlated ID

Example of Log display window

5.5 UMS Security management

◆ Access control of user

- UMS provides three authority levels for user security.
 - The highest level of user can monitor all information, operate UMS system function such as Database backup, and so on.
 - The lowest level of user can monitor some information.
- Login user is authenticated with password.
 - User ID and Password is stored in UMS Database.
- These settings of user security are set in User information window.

UMS User Information

Operation: ☒ Add User ☐ Delete User ☐ Modify User

Username	PIC	PSD
hogeuser		
hogeuser3	✓	✓

User Information

User Name:

Password:

Retype Password:

User Level:

Security Setting:

WLD operator:

IP Address Filter

IP Address:

MAC Address Filter

MAC Address:

Login Management

UMS Synchronize Mode

☐ SYNC ☐ ASYNC

Add User Cancel

Example of User information window

Site User Configuration - hogeuser3

UMS Username: hogeuser3

Site Name	Status
hogeuser	Enable SSO
hogeuser2	Enable SSO
hogeuser3	Enable SSO
hogeuser4	Enable SSO
hogeuser5	Enable SSO
hogeuser6	Enable SSO
hogeuser7	Enable SSO
hogeuser8	Enable SSO
hogeuser9	Enable SSO
hogeuser10	Enable SSO
hogeuser11	Enable SSO
hogeuser12	Enable SSO
hogeuser13	Enable SSO
hogeuser14	Enable SSO
hogeuser15	Enable SSO

User Setting

User Level:

☐ Disable Single Sign On

☐ Enable Single Sign On

☐ Disable Login

IP Address Filter

IP Address:

MAC Address Filter

MAC Address:

UMS Synchronize Mode

☐ SYNC ☐ ASYNC

Apply Cancel

Example of User Configuration window

5.5 UMS Security management

◆ Security Setting with FP unit and Function unit

- UMS provides FP unit security.
- UMS also provides the security for the system control function.
 - The permission for the system control such as “power feeding path switching” and “optical path control” can be set.
- These security settings are set in Security Setting window, and can be set to each user individually.

Name	Long form	Authority
PIC	Power Indication Coordinator	To control Power Feeding Path.
PSO	Power Safety Officer	To join the operation of Power Feeding Path controlling according to the instructions by PIC user.
OMO	Optical Management Operator	To control Optical Path.(ROADM and Optical SW in BU)

ABC NOC UMS Security Setting

Login User Information

User Name test

FP

FP name

☒ FP01

☒ FP02

☒ FP03

☒ FP04

☒ FP05

☒ FP06

☒ FP07

☒ FP08

☒ FP09

☒ FP10

☒ FP11

☒ FP12

☒ FP13

☒ FP14

☒ FP15

☒ FP16

Select All Remove All

Function

Function name

☒ Power Feeding Path

☒ PIC ☒ PSO

☒ DWA Path

☒ ROADM(WSS) Control (OMO) / Optical SW

NE

NE name

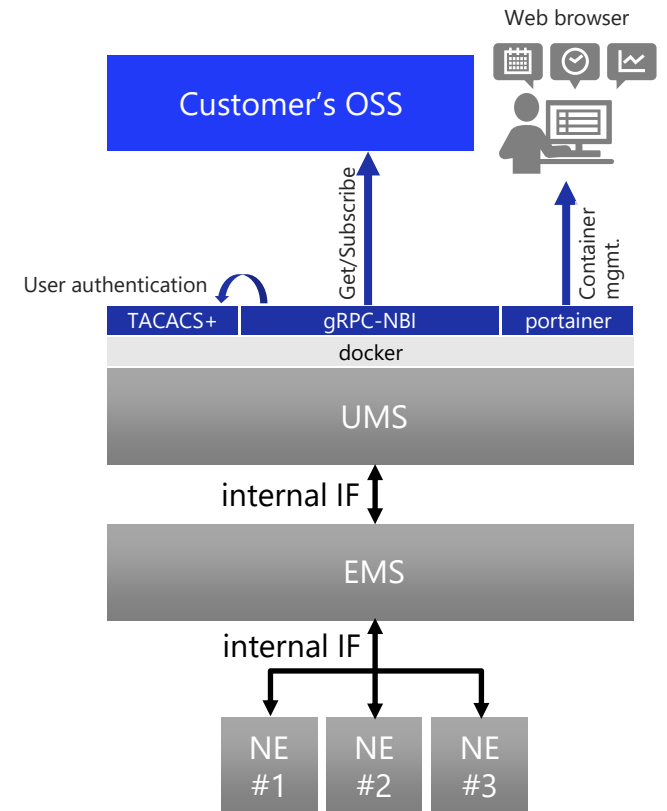
☒ PFE

Select All Remove All Apply Cancel

Example of Security Setting window

5.6 UMS NBI (North-Bound Interface)

- ◆ NEC's gRPC-based north-bound interface (NBI) at UMS is capable of exposing alarms to customer's operation support system (OSS).
- ◆ APIs for gRPC-NBI
 - Get/Subscribe alarms
 - Get inventory
 - Get/Subscribe PMs (OCI [including RSM] / PFE / TPND)
- ◆ User authentication using TACACS+
- ◆ SSL-based secured gRPC channels
- ◆ Logical Partitioning
 - Maintaining confidentiality of cable resources owned by other customers
- ◆ Portainer is a container management tool, which enables operators to check and manage status of NBI app.



5.6 UMS NBI (North-Bound Interface)

◆ APIs for gRPC-NBI, defined by proto files

- **Get(type: unary) / Subscribe (type: server streaming) alarms**
- Get inventory
- Get/Subscribe PMs (OCI [including RSM] / PFE / TPND)

PB service	RPC name	Type	Description
Alarm	GetCurrentAlarm	Unary	Retrieve the current active alarms on the alarm journal
	AlarmSubscription	Server streaming	Subscribe the alarms, and then get notified when an alarm is raised or cleared
PM	GetMonitorData	Unary	Retrieve the performance monitor data of OCI (including RFTE) / PFE / TPND stored in the UMS server
	MonitorSubscription	Server streaming	Subscribe the PM update notification, and then get notified when monitor information is updated
Inventory	GetInventoryData	Unary	Retrieve the inventory information, including serial number, of OCI (including RFTE) / PFE / TPND
	GetStationList	Unary	Retrieve the landing station list and NOC list
Login logs	GetLogs	Unary	Retrieve the gRPC-NBI login logs

5.6 UMS NBI (North-Bound Interface)

- ◆ Alarm events which also appear in the Alarm Journal window at UMS are exposed through gRPC-NBI.
 - Flatbox OCI (BIF, FDL, RBC, FPS, FSA20, RFTE)
 - PFE
 - Wet-plant equipment (WSS-ROADM)
 - EMS
 - UMS

6. Qualification Status

6.1 Qualification Status of WebNSV EMS

6.2 Qualification Status of WebNSV UMS

6.1 Qualification Status of WebNSV EMS

◆ Qualification completed in June 2022.

No	Qualification Test Items	Result
1	Basic Function	Pass
2	Fault Management Function	Pass
3	Performance Management Function	Pass
4	Configuration Management Function	Pass
5	Log Management Function	Pass
6	Security Management Function	Pass

6.2 Qualification Status of WebNSV UMS

◆ Qualification completed in June 2022.

No	Qualification Test Items	Result
1	Basic Function	Pass
2	Fault Management Function	Pass
3	Performance Management Function	Pass
4	Configuration Management Function	Pass
5	Log Management Function	Pass
6	Security Management Function	Pass
7	NBI Function	Pass

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Appendix: UMS NBI (North-Bound Interface)

◆ Alarm message format

Entity name	Datatype	Description
severity	String	Alarm severity (critical / major / minor / warning / indeterminate / cleared / unknown)
notificationId	Long	Notification ID / Cleared ID
correlatedId	Long	Correlated notification ID
userLabel	String	User label of the NE raising an alarm
siteName	String	Site name of the NE raising an alarm
probableCause	String	Probable cause of the NE raising an alarm
additionalText	String	Additional text of the NE raising an alarm
eventTime	String	Time when an alarm is raised
fp_number	String	FP information associated with the equipment
instanceName	String	Instance name of the NE raising an alarm
lineInstanceName	String	Line instance name of the NE raising an alarm

Appendix: UMS NBI (North-Bound Interface)

◆ Alarm message example

```
{
  "grpcAlarmNotificationStream": {
    "severity": "MAJOR",
    "notificationId": "52",
    "correlatedId": "0",
    "siteName": "StationB",
    "instanceName": "/Site/EQP/RACK8/Rack8/TPNDFShelf322/TPNDF00010100002/MuxPonder_S03/LINE1",
    "userLabel": "TPND322STB_5133_5121/S03_XF1200/LINE1",
    "probableCause": "aIS",
    "additionalText": "LINE OUT(AMS1(CH1))",
    "lineInstanceName": "/Site/EQP/RACK8/Rack8/TPNDFShelf322/TPNDF00010100002/MuxPonder_S03/LINE1",
    "eventTime": "Thu Aug 17 05:26:56 UTC 2023",
  },
  "response": {
    "code": 0,
    "message": "Success"
  }
}
```

Appendix: UMS NBI (North-Bound Interface)

◆ Alarm message example

```
{
  "grpcAlarmNotificationStream": {
    "severity": "CLEARED",
    "notificationId": "53",
    "correlatedId": "52",
    "siteName": "StationB",
    "instanceName": "/Site/EQP/RACK8/Rack8/TPNDFShelf322/TPNDF00010100002/MuxPonder_S03/LINE1",
    "userLabel": "TPND322STB_5133_5121/S03_XF1200/LINE1",
    "probableCause": "aIS",
    "additionalText": "LINE OUT(AMS1(CH1))",
    "lineInstanceName": "/Site/EQP/RACK8/Rack8/TPNDFShelf322/TPNDF00010100002/MuxPonder_S03/LINE1",
    "eventTime": "Thu Aug 17 05:26:56 UTC 2023",
  },
  "response": {
    "code": 0,
    "message": "Success"
  }
}
```